

Product Requirements Document (PRD)

Project Name

GymGear Marketplace

Version

V1 MVP

Overview

GymGear Marketplace is a B2C + C2C e-commerce platform focused on gym and fitness equipment. Users can buy and sell new or used gym products through a unified marketplace.

The platform will support:

- Buyer accounts
 - Seller accounts
 - Admin accounts
 - Product listing and management
 - Product search and filtering
 - Cart and checkout
 - Cash on Delivery orders
 - Reviews and ratings
 - Buyer-seller messaging
 - Administrative moderation
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Objectives

Business Goals

- Create a niche marketplace for fitness equipment.
- Enable individuals and businesses to sell equipment.
- Provide a simple buying experience.
- Launch MVP within a short timeline.

User Goals

Buyers should be able to:

- Search products

- Compare products
- Contact sellers
- Purchase equipment

Sellers should be able to:

- Create listings
- Manage inventory
- Track orders
- Communicate with buyers

Admins should be able to:

- Moderate products
 - Manage users
 - Monitor platform activity
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Tech Stack

Frontend

- React
- React Router
- Tailwind CSS
- Axios
- Context API

Backend

- Node.js
- Express.js

Database

- MongoDB
- Mongoose

Authentication

- Clerk

Real-Time Communication

- Socket.io

Image Storage

- MongoDB (GridFS)

Payment

- Cash on Delivery (COD)
-

User Roles

Buyer

Permissions:

- View products
- Search products
- Add items to cart
- Place orders
- Chat with sellers
- Leave reviews

Seller

Permissions:

- Create products
- Edit products
- Delete products
- View orders
- Chat with buyers
- Manage inventory

Admin

Permissions:

- Manage users
 - Manage products
 - Remove listings
 - View reports
 - Moderate reviews
-

Core Features

Authentication

Registration

Users can register through Clerk.

Fields:

- Full Name
- Email
- Password

Login

Users can log in using Clerk authentication.

User Profile

Fields:

- Name
- Email
- Profile Picture
- Role
- Location

Product Management

Create Product

Seller can add:

- Product Name
- Description
- Category
- Condition (New/Used)
- Price
- Quantity
- Images
- Location

Edit Product

Seller can update:

- Price
- Quantity
- Description
- Images

Delete Product

Soft delete recommended.

Product Categories

- Dumbbells
 - Barbells
 - Weight Plates
 - Kettlebells
 - Benches
 - Treadmills
 - Exercise Bikes
 - Resistance Bands
 - Home Gym Systems
 - Accessories
-

Product Search

Features:

- Search by keyword
 - Filter by category
 - Filter by price
 - Filter by condition
 - Sort by newest
 - Sort by price low-high
 - Sort by price high-low
-

Product Details Page

Displays:

- Product Images

- Product Name
- Price
- Description
- Seller Information
- Reviews
- Rating
- Stock Status

Actions:

- Add to Cart
 - Contact Seller
-

Cart

Features:

- Add Item
 - Remove Item
 - Update Quantity
 - View Total
-

Checkout

Flow:

Cart → Address → Order Summary → COD Confirmation → Order Created

Orders

Buyer

Can:

- View Orders
- Track Status

Statuses:

- Pending
- Confirmed
- Shipped
- Delivered

- Cancelled

Seller

Can:

- View Incoming Orders
 - Update Status
-

Reviews & Ratings

Buyers can submit:

- Rating (1-5)
- Comment

Rules:

- Review only after delivery
 - One review per order
-

Chat System

Buyer-Seller Messaging

Features:

- Send Message
- Receive Message
- View Conversation History

Technology:

- Socket.io
-

Seller Dashboard

Overview

Displays:

- Total Products
- Active Listings

- Total Orders
- Revenue

Product Management

- Add Product
- Edit Product
- Delete Product

Order Management

- View Orders
 - Update Status
-

Admin Dashboard

User Management

Admin can:

- View Users
- Suspend Users
- Delete Users

Product Moderation

Admin can:

- Approve Listings
- Reject Listings
- Remove Listings

Reports

Admin can view:

- Total Users
 - Total Sellers
 - Total Orders
 - Total Products
-

Database Design

Users Collection

```
{
  _id,
  clerkId,
  name,
  email,
  role,
  avatar,
  createdAt
}
```

Products Collection

```
{
  _id,
  sellerId,
  title,
  description,
  category,
  condition,
  price,
  quantity,
  images,
  status,
  createdAt
}
```

Cart Collection

```
{
  _id,
  userId,
  items: [
    {
      productId,
      quantity
    }
  ]
}
```

```
]
}
```

Orders Collection

```
{
  _id,
  buyerId,
  sellerId,
  products,
  totalPrice,
  shippingAddress,
  status,
  createdAt
}
```

Reviews Collection

```
{
  _id,
  productId,
  userId,
  rating,
  comment,
  createdAt
}
```

Messages Collection

```
{
  _id,
  senderId,
  receiverId,
  message,
  createdAt
}
```

API Modules

Auth

- POST /sync-user

Products

- GET /products
- GET /products/:id
- POST /products
- PUT /products/:id
- DELETE /products/:id

Cart

- GET /cart
- POST /cart/add
- DELETE /cart/remove

Orders

- POST /orders
- GET /orders
- PUT /orders/:id/status

Reviews

- POST /reviews
- GET /reviews/:productId

Chat

- GET /messages
- POST /messages

Admin

- GET /users
 - DELETE /users/:id
 - GET /reports
-

Success Metrics

Launch Goals:

- User Registration
 - Product Listing Creation
 - Successful Orders
 - Review Submission
 - Active Seller Usage
 - Daily Product Uploads
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Future Features (V2)

- Wishlist
- AI Product Recommendations
- Razorpay Integration
- Location-Based Search
- Product Approval Workflow
- Seller Verification
- Analytics Dashboard
- Notifications
- Email Alerts
- Mobile App
- Multi-language Support